

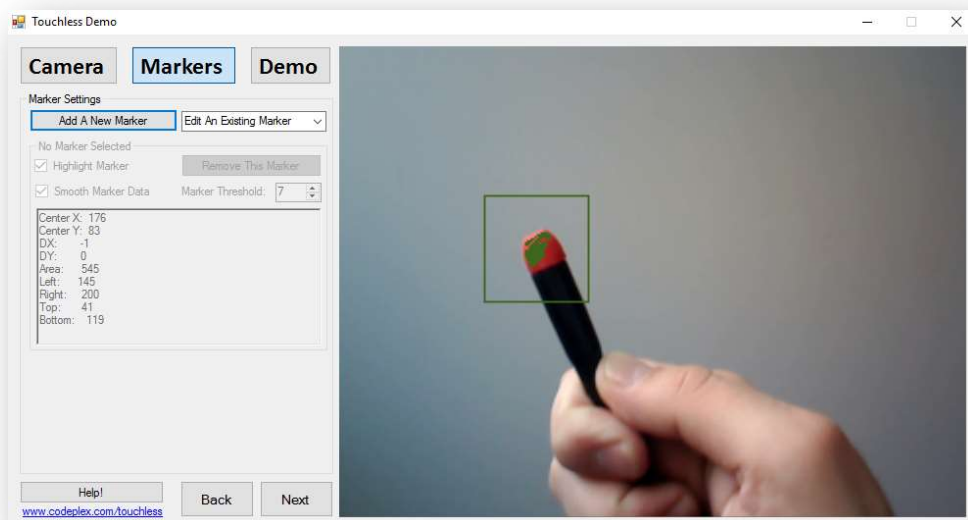
Human-Computer Interaction

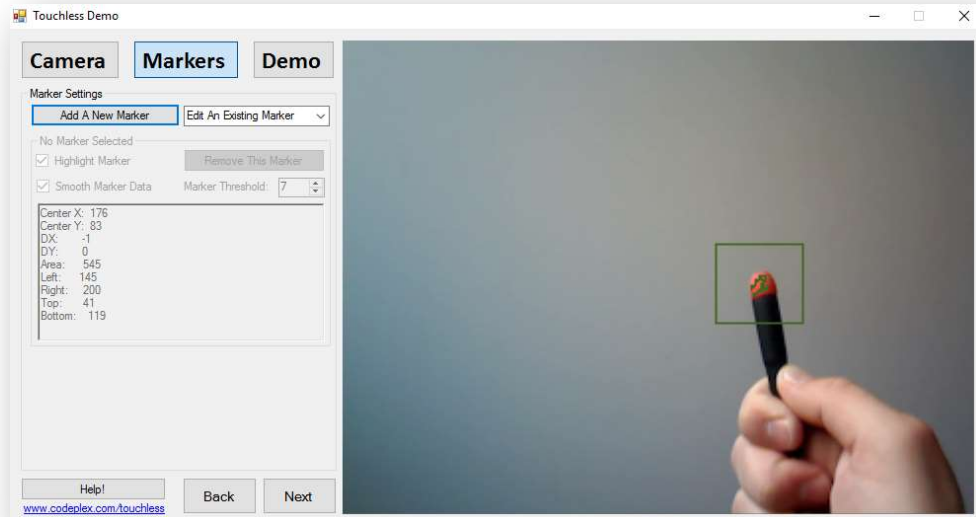
Camera mouse

- The aim of the exercise is to become familiar with the possibilities of controlling the pointer (mouse) on the screen using a gesture – in a touchless mode – utilizing a webcam.
- One of the simplest mechanisms are marker-based systems. They rely on an object with a distinctive color. It is detected and tracked, and changes in its position are mapped to cursor movement.

Exercise 1 - Example solution

- Check out the example software for detecting and tracking objects using color provided in the zip file.
- The basis for control is the color-distinctive object in the scene (marker). Take any object (you have at hand) and using the camera preview add a new marker that will be used for steering.
- Verify if the system effectively tracks the object as we move it in the air. In case of problems, use a different object or improve the lighting. Keep in mind that the object should be unique in color (stand out from the scene), and the scene should be well-lit.
- Examples:





- In the *Demo* tab, there is an option to assess the effectiveness of our indicator in the touchless interface control. Several examples are available, such as the *Snake* game. Check each example (some may require adding more than one object to the control). Test the effectiveness of the interface. Your conclusions should be included in the report.

Exercise 2 - testing what factors influence the detection of a marker

- Carry out experiments in which you will check the effectiveness of tracking an object (marker) depending on its color, shape, size, distance from the camera, background complexity (scene), speed of movement and lighting condition. Document the results in the report.
- Tip: Try to also test a hand (or your face) as one of the markers. Is the color detection mechanism used in the evaluated solution effective in this case?

Report

- This is the first part and will be continued.
- Begin preparing the report on the exercises performed, in the PDF format.
- Don't send for now.